QuPath

QuPath - Open Source Digital Pathology

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Documentation

Source code

QuPath is open source software for **Quantitative Pathology**

QuPath aims to help improve the speed, objectivity and reproducibility of digital pathology analysis and biomarker interpretation by offering an open, powerful, flexible, extensible software platform for whole slide image analysis.

QuPath has been developed for research applications at the Centre for Cancer Research & Cell Biology at Queen's University Belfast, as part of research projects funded by Invest Northern Ireland and Cancer Research UK.

Latest news

QuPath publications now online!

The primary publication describing QuPath is now available open access - please cite it if you use QuPath in your work!

Bankhead, P. et al.
QuPath: Open source software for digital pathology image analysis.
Sci. Rep. 7, 16878 (2017)

Other publications involving QuPath are listed in the documentation.

Better support for more whole slide image formats

The QuPath Bio-Formats extension received a major update in January 2018. Lots of bug fixes and performance improvements now mean that more whole slide formats work better and faster with QuPath.

Read more about the update here.

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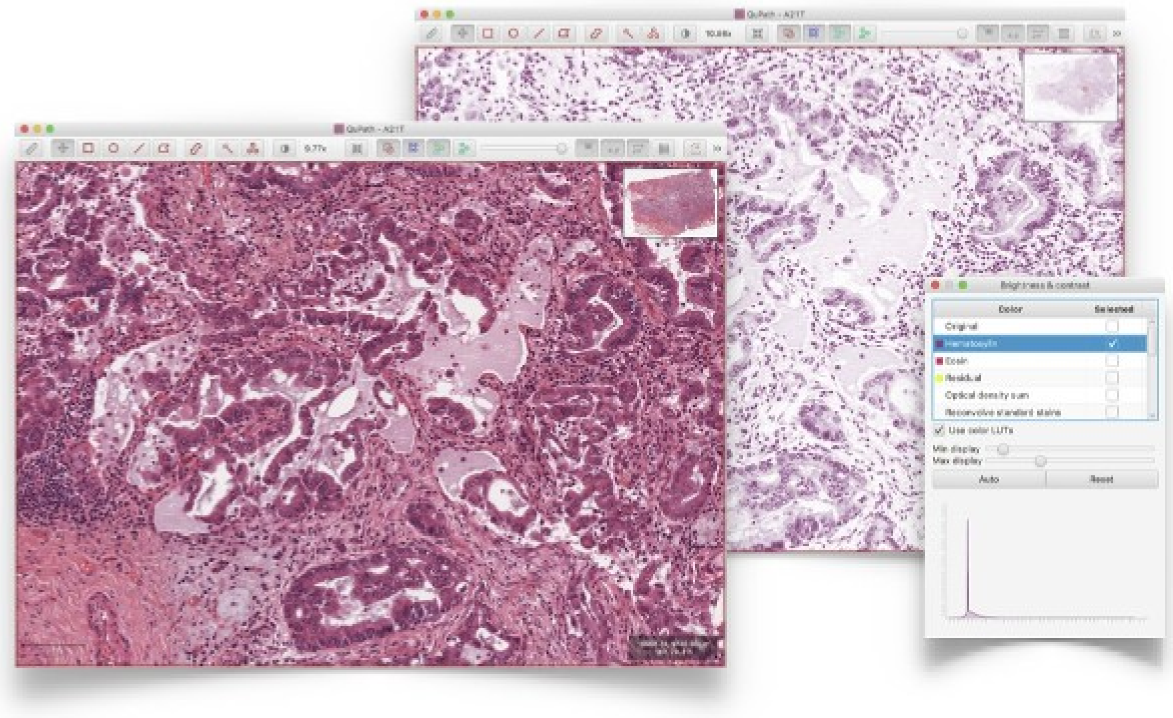
More news coming soon...

In the meantime, don't forget there's a QuPath forum at Google Groups, where questions are asked, ideas are shared, and announcements are made.

Features

Whole slide viewing



Fast, flexible image viewer capable of displaying whole slide images (often > 30 GB uncompressed) using dynamic color transforms (e.g. stain separation) and tracking slide navigation

Biomarker quantification

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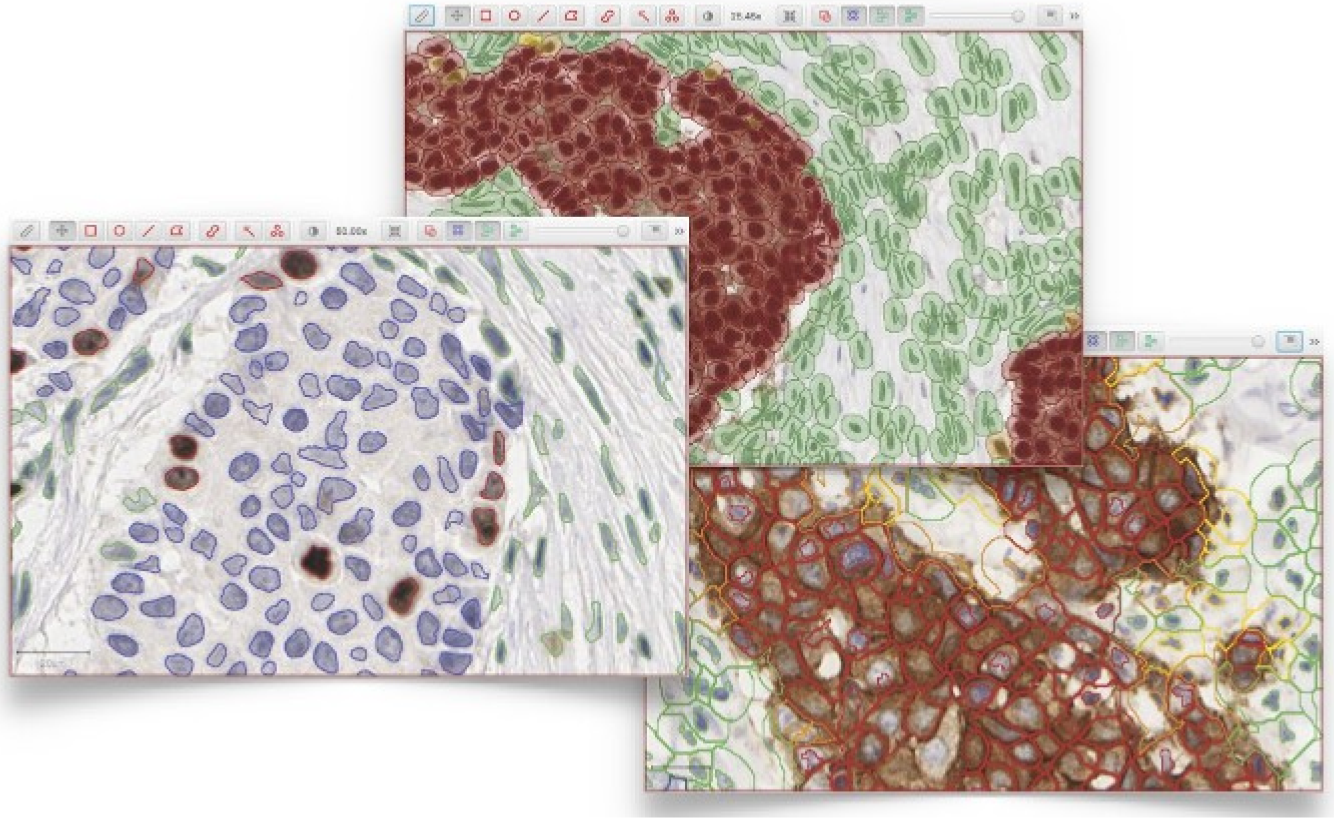
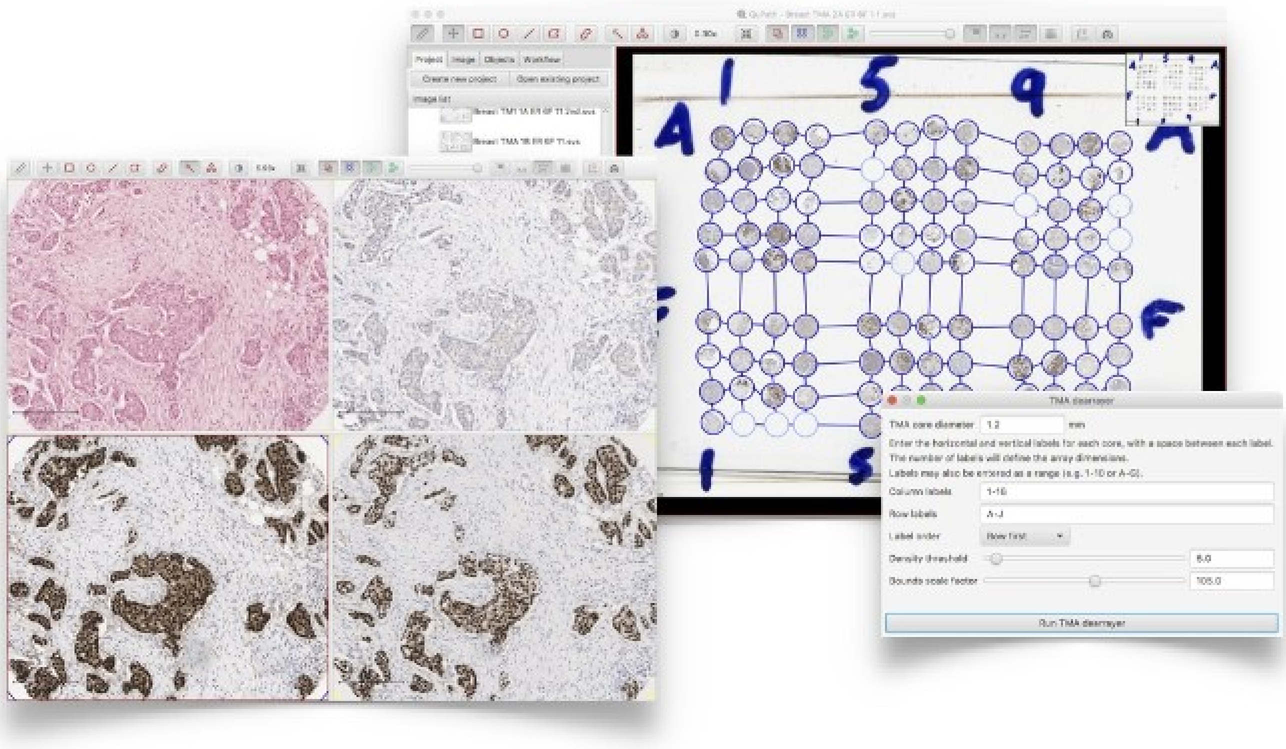


Nuclear, cytoplasmic & membranous biomarkers can all be quantified quickly using automated segmentation algorithms combined with trainable cell classification

Tissue Microarray support



Automated dearraying of Tissue Microarrays and the ability to view related cores side-by-side

Sophisticated tumor identification

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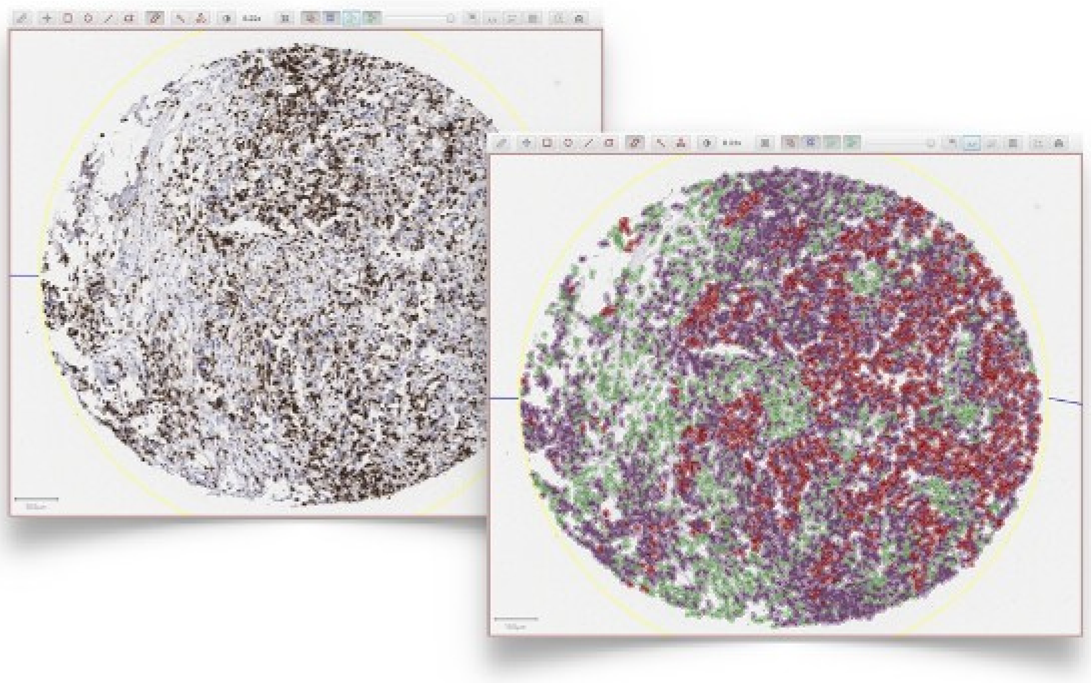
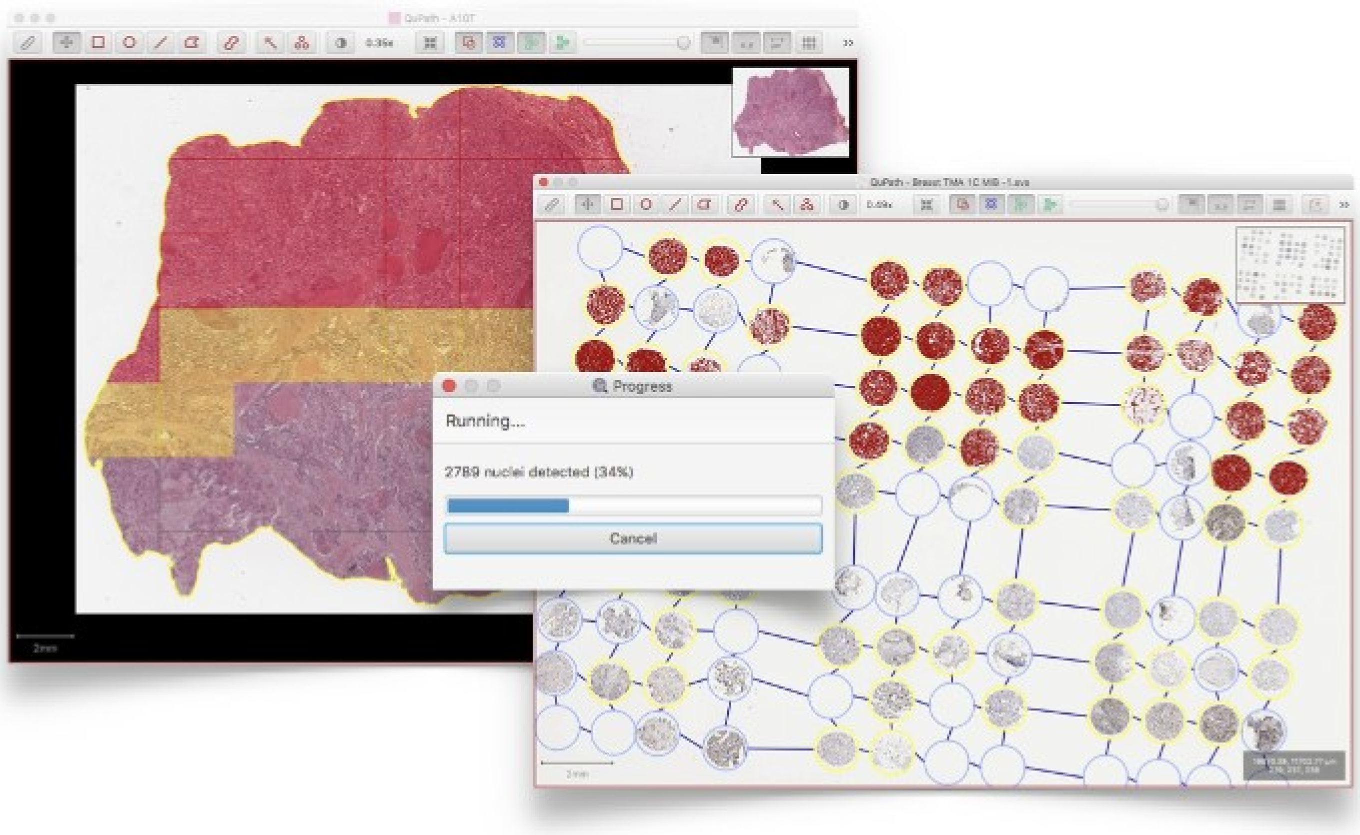


Powerful tumor identification algorithms can be applied directly to slides of interest - including slides stained for immune cells - without the need to stain for a separate tumor marker

Fast analysis



Large image regions are split into tiles where necessary, and these tiles analyzed in parallel with efficient algorithms - giving fast results without requiring specialist hardware

Flexible object classification

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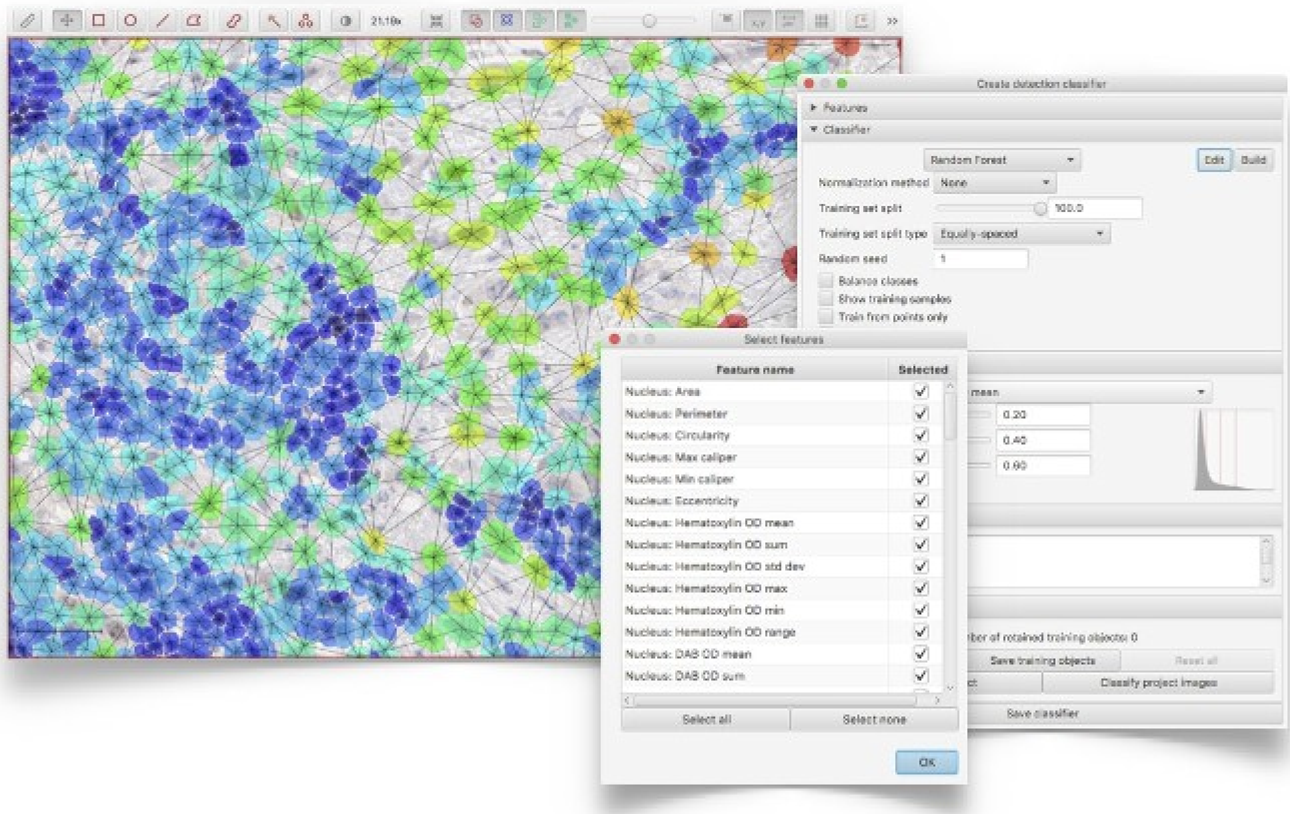
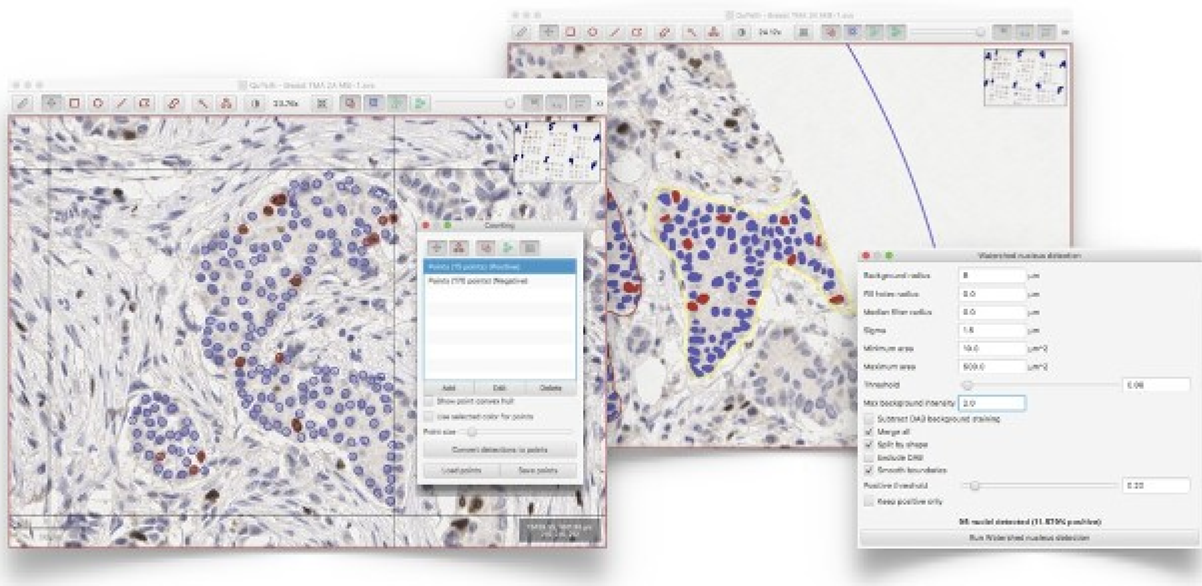


Apply object classification with the default ‘out-of-the-box’ random forest classifier (from OpenCV), or create highly-customized algorithms by tuning the choice of classifier, parameters and features used

Interactive tools



Extensive tools for slide navigation, annotating areas, exporting image regions or counting cells - either manually, or using automated cell detection

User-friendly automated analysis

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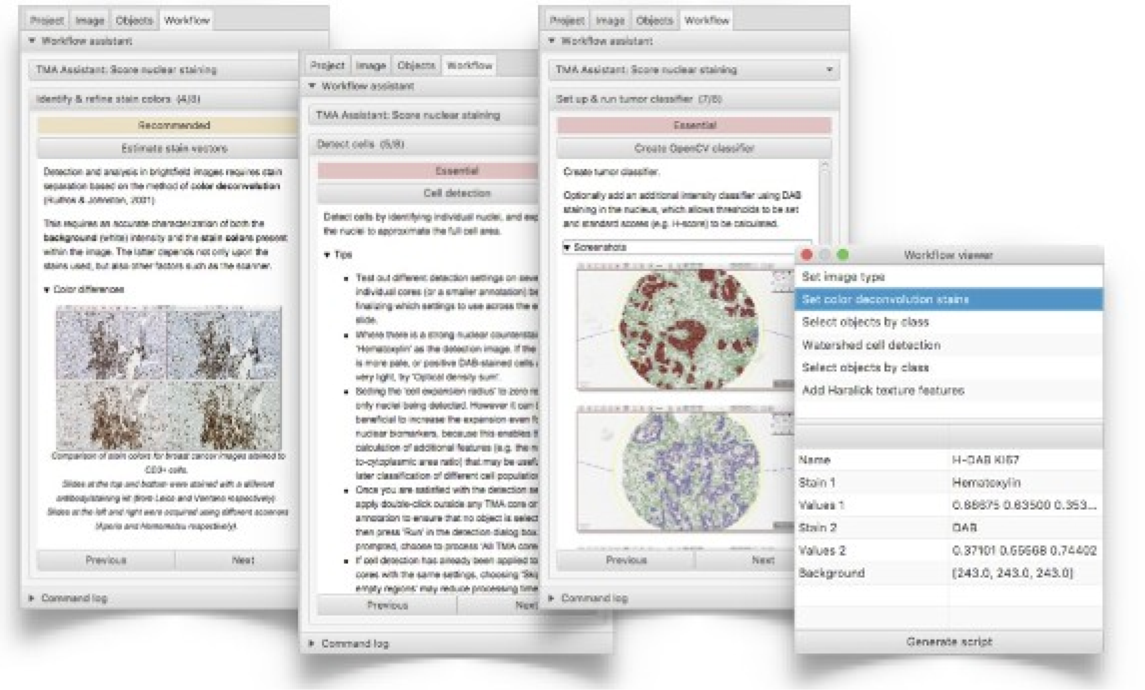
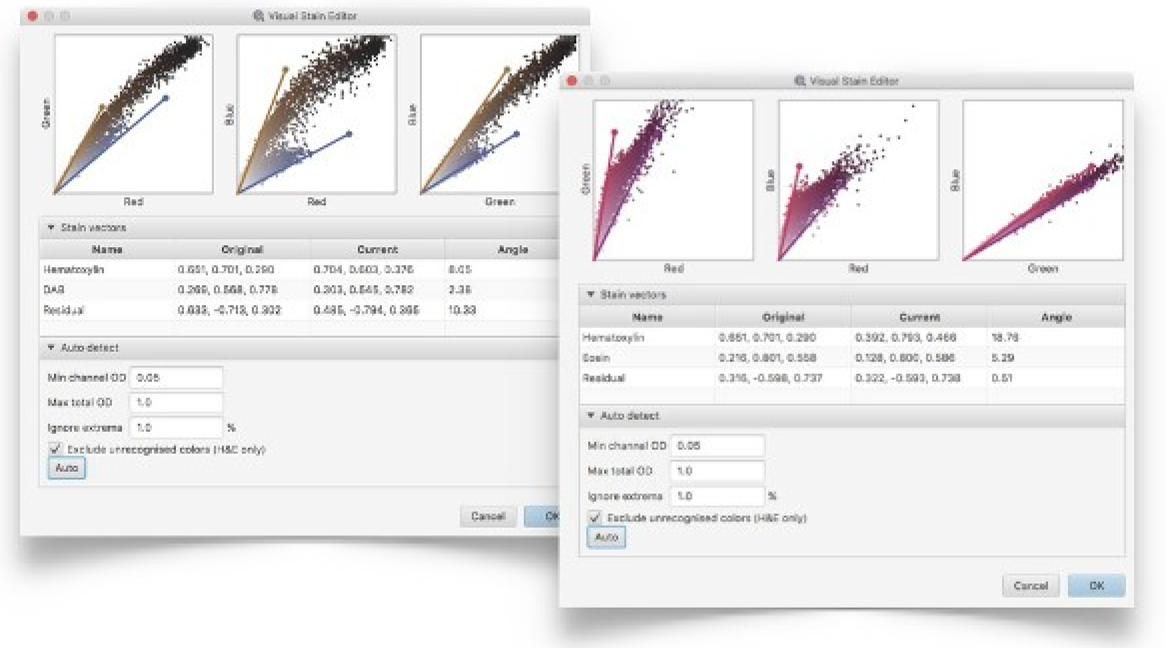


Workflows provide guided analysis for common tasks, or users can devise their own approaches by running commands in any order, which are automatically logged for reproducibility

Stain estimation



Analysis can be tailored to different stains and scanners using advanced stain estimation, visualization & optimization tools

Scripting

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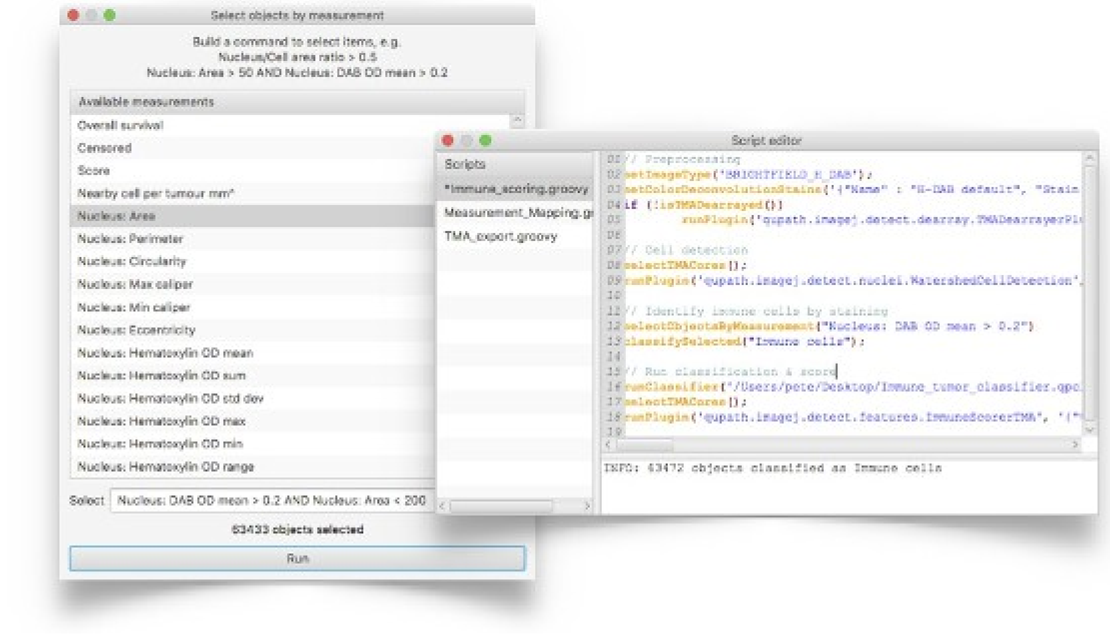
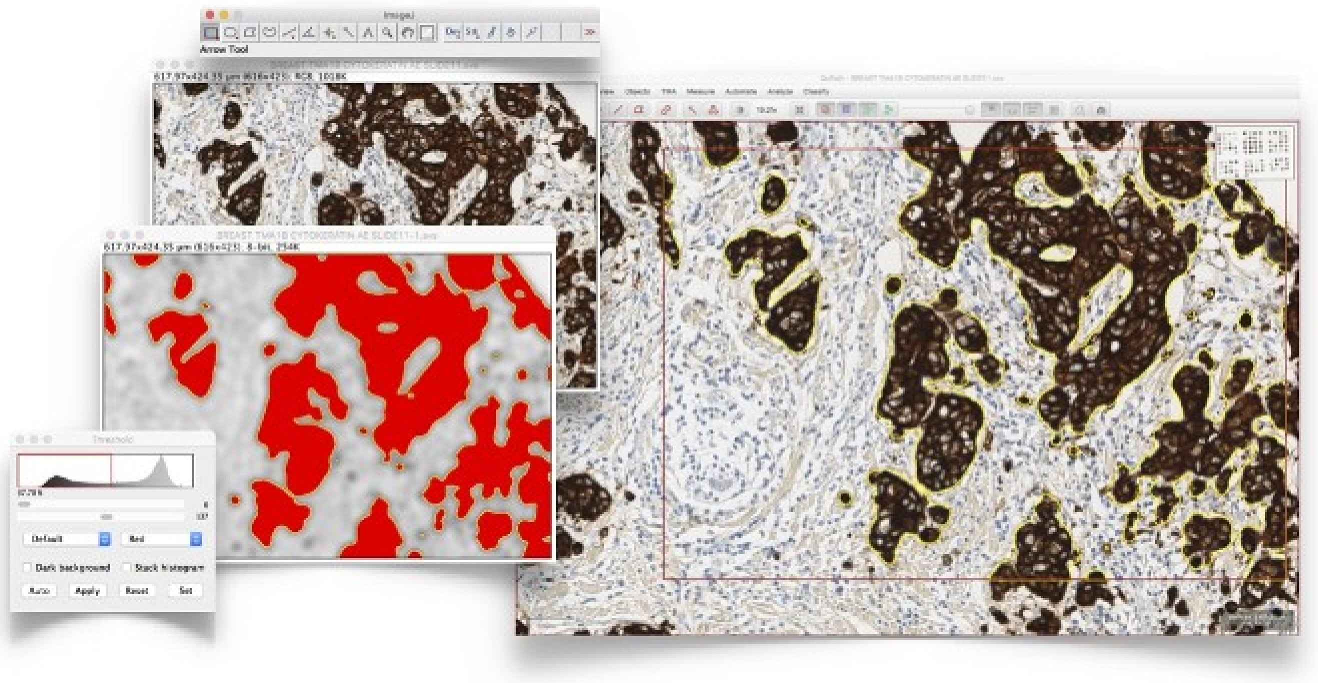


Experienced users can enter commands and write scripts to perform sophisticated, customized analysis using QuPath’s powerful, efficient hierarchical data structures

Data exchange



Exchange data with open source tools (e.g. ImageJ), or read images from a variety of sources, including cloud-based hosting

Analytics & export

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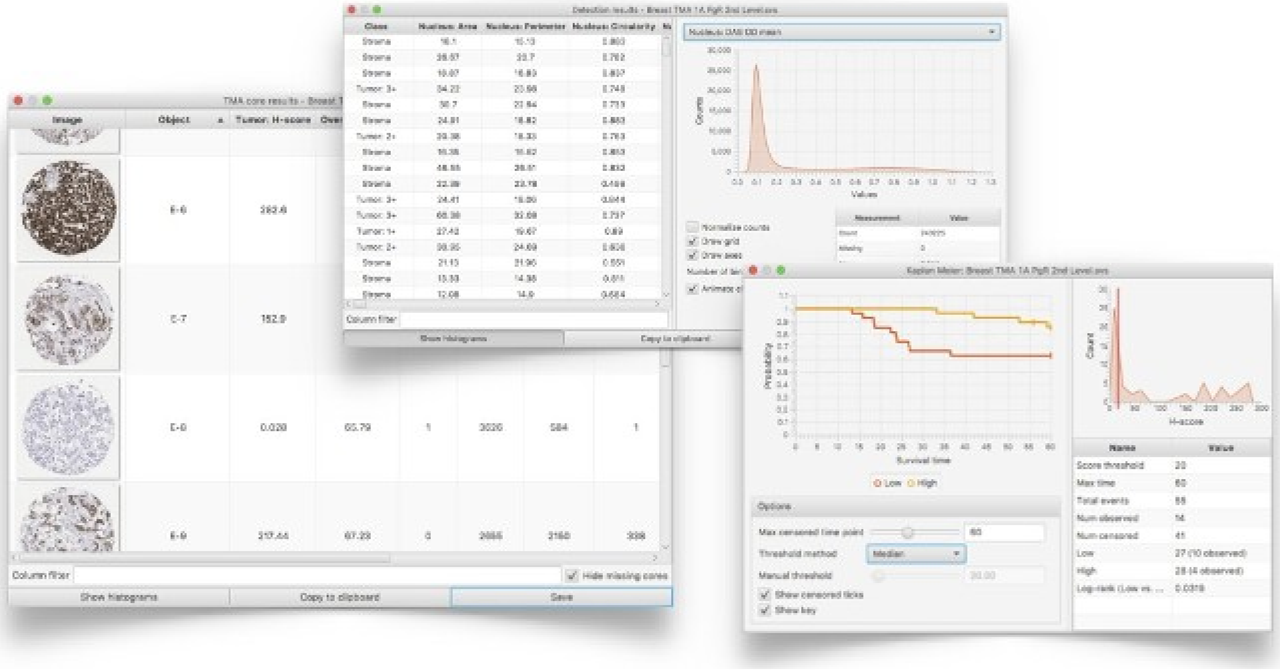
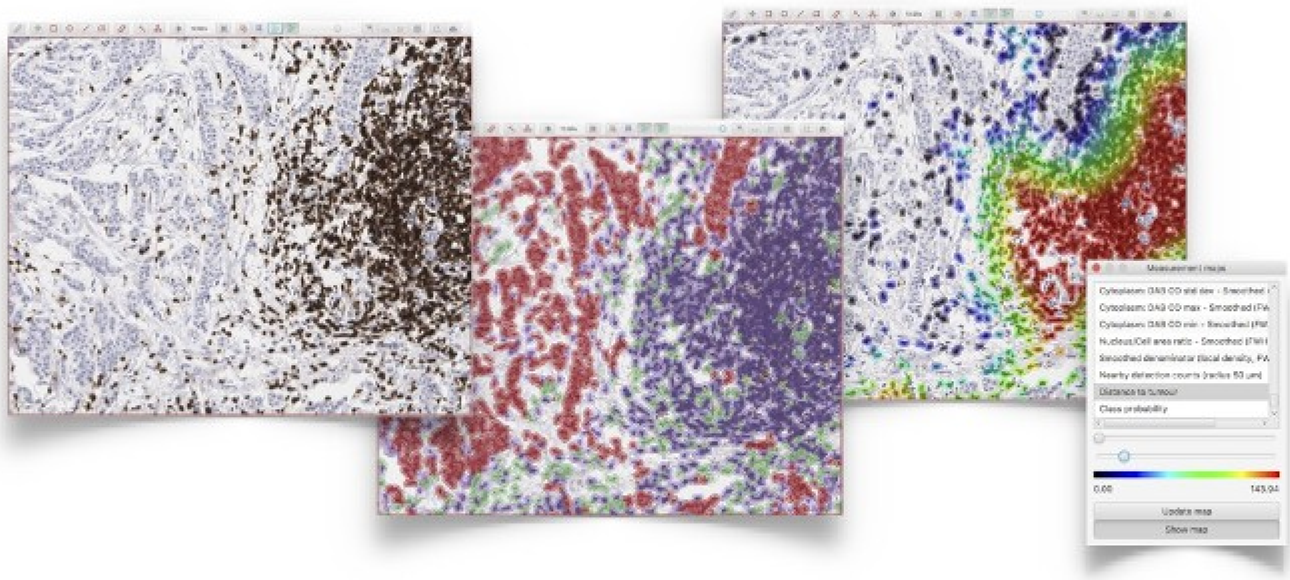


Create interactive results tables, histograms, scatterplots & survival curves directly within QuPath, or export results in standard formats to import into other software if required

Visualization



View measurements in context by color coding objects according to their features, e.g. to identify hotspots or visualize cell distributions for immuno-oncology applications

Versatility

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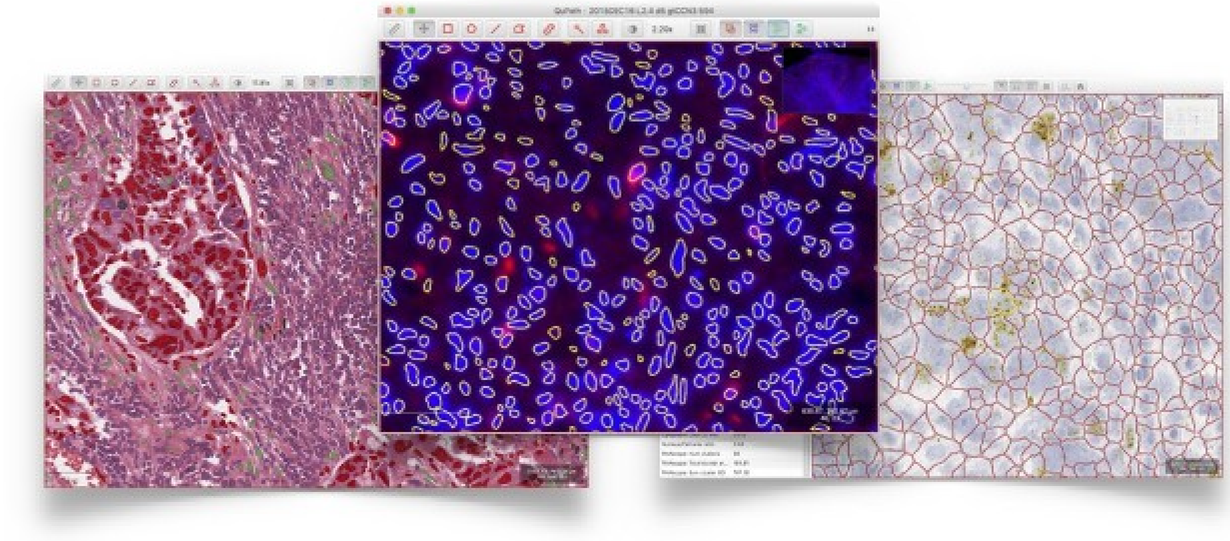

QuPath has been developed as a cross-platform application that runs on Windows, Mac OS X and Linux to support a wide range of applications and image types across pathology and the biosciences